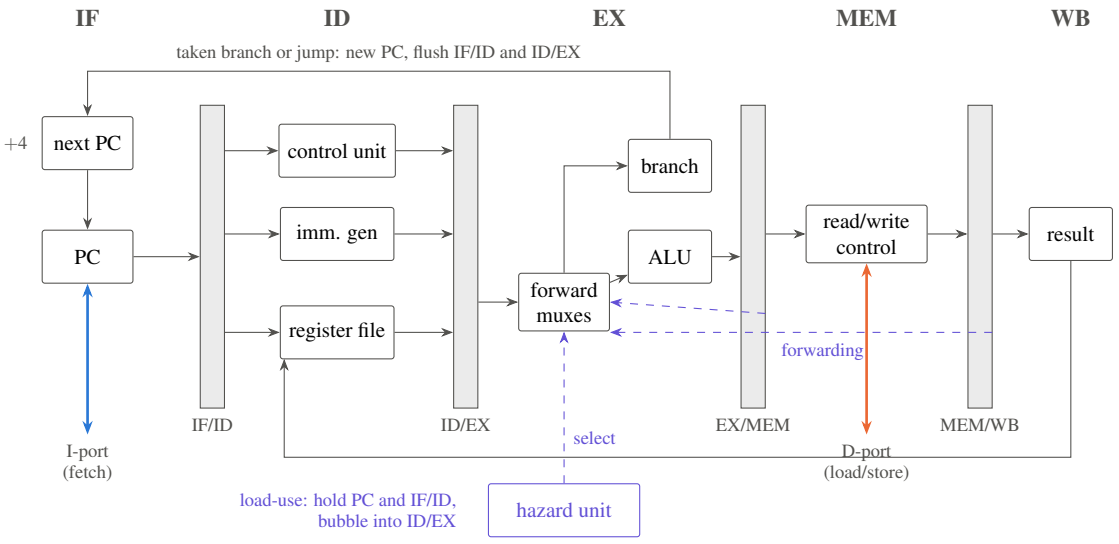

fetch not ready → bubble into ID data port not ready → IF to MEM freeze, bubble into WB every empty slot records why it is empty